

EduGenie: Google Gemini Powered Learning Assistant

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Project Description

EduGenie is a Google Gemini powered learning assistant designed to make student learning simple, personalized and confidence-building. It supports explanation, question answering, summarization, quiz generation and learning recommendations through one student-friendly web interface.

The application accepts a topic, question or study passage and routes the request through a FastAPI backend. Depending on the selected task, it uses a local LaMini-Flan-T5 model for explanation when enabled, or Google Gemini for Q&A, quizzes, summaries and learning recommendations.

Core Purpose

- Understand difficult concepts in simple language.
- Clear doubts quickly.
- Turn long material into concise revision points.
- Generate personalized practice questions.
- Create a logical learning path and suggest what to study next.

Main Features

- Explain a concept
- Ask a question
- Generate a three-question MCQ quiz
- Summarize notes
- Create a learning path
- Learning-level selection: beginner, intermediate and advanced
- Interactive quiz feedback and copyable results

1. IDEATION PHASE – Problem Statements

EduGenie – Problem Statements

PS-1: Understanding & Clearing Doubts

Students find difficult topics hard to understand. Online explanations can be too technical. Solution: simple, student-friendly AI explanations.

PS-2: Knowing What to Learn Next

Students are confused about what to learn first and next. Learning resources are not always organized. Solution: personalized learning roadmap.

PS-3: Revision & Practice

Revising long study materials takes time. Students may not have ready-made practice questions. Solution: concise revision points and personalized practice questions.

Ideation Phase – Focus

- Identify student learning problems.
- Generate different solution ideas.
- Group ideas based on common purposes.
- Prioritize ideas by importance and feasibility.

Core Learning Need

EduGenie brings explanation, planning, revision and practice together so the student can move from a doubt to a structured next step.

Ideation – Brainstorming, Idea Groups & Capability Mapping

1. Brainstorming

- Identify student learning problems.
- Generate different solution ideas.
- Group ideas based on common purposes.
- Prioritize ideas by importance and feasibility.

2. Core Problem Statements

- PS-1: Students may not know their learning gaps.
- PS-2: Students struggle to organize study time.
- PS-3: Students find it difficult to track progress and maintain learning context.

3. Five Main Idea Groups

1. Adaptive Learning Diagnosis
2. Smart Study Planning
3. Multimodal Learning Representation
4. Progress Analytics & Goal Tracking
5. Context-Aware Academic Assistance

4. Capability Mapping

Problem → Student Need → Idea Group → EduGenie Capability. EduGenie converts identified learning problems into practical AI-based capabilities.

Problem	Student Need	Idea Group	EduGenie Capability
Difficult concepts	Simple explanation	Adaptive Learning Diagnosis	Level-aware concept explanation
Unclear next topic	Ordered plan	Smart Study Planning	Learning roadmap
Long material	Fast revision	Multimodal Learning Representation	Summary and revision points
Weak progress visibility	Track goals	Progress Analytics & Goal Tracking	Progress-oriented support
Generic answers	Relevant context	Context-Aware Academic Assistance	Context-aware chat

5. Idea Prioritization

Ideas are compared using Importance vs Feasibility. No numerical scoring is used; ideas are considered based on usefulness and implementation practicality.

6. Final Outcome

- Understand the learner.
- Identify learning gaps.
- Create study plans.
- Present content in different formats.
- Track learning progress.
- Provide context-aware academic support.
- Give concise revision points and personalized practice questions.

Result

The ideation phase connects real student problems with practical AI capabilities and establishes the basis for requirements analysis.

Empathy Map

User Persona

School/college student using EduGenie to learn, plan and revise.

Think & Feel

- Will I understand this topic?
- Feeling confused, low confidence.
- Am I studying the right things?
- Worried about doing badly in exams.

Say & Do

- Says: "I don't understand this topic."
- Searches the same doubt on many websites.
- Delays revision and practice until exam time.
- Makes timetables but can't follow them.

Hear

- Friends: "Just watch videos online."
- Teachers: "Revise regularly and practise."
- Parents: "Focus on your studies."
- Everyone studies differently.

See

- Overly technical online explanations.
- Study resources scattered everywhere.
- Long study materials to revise again.
- Classmates who seem better prepared.

Pains

- Explanations too difficult for my level.
- Don't know what to learn first or next.
- Revision and making practice questions take too long.
- Unsure of understanding before exams.

Gains

- Simple student-friendly explanations.
- Personalized learning roadmap.
- Concise revision points and practice questions.
- More confidence and better exam preparation.

Student Learning Journey

7. Entice

Student struggles with a concept → notices a knowledge gap → decides to try EduGenie.

8. Enter

Identifies need: explain, ask, summarize, practice or plan → opens EduGenie → picks a task.

9. Engage

Types concept, doubt, long passage or topic → submits request → result appears below input.

10. Exit

Reads explanation/answer → reviews revision points → takes quiz → reads learning roadmap.

11. Extend

Sees what to study next → continues next topic or revises notes → returns with new doubts.

Main Goals

- Understand simply.
- Clear doubts.
- Revise quickly.
- Practice.
- Follow topics logically.
- Know what to learn next.

Positive Moments

- One page, many study helps.
- Single dropdown.
- Simple explanations.
- Long content becomes short points.
- Wrong quiz answers corrected.
- Roadmap shows where to start.

Learning Journey – Negative Moments & Opportunities

Negative Moments

- Unsure which help type to choose.
- Hard to word the question.
- Too little context may give weak answers.
- Output may need checking.
- Deeper doubts may need a teacher.
- Three questions may not cover the whole topic.
- Path may not match learner level.

Areas of Opportunity

- Quick diagnostic.
- Hints for choosing tasks.
- Sample questions.
- Subject/course context.
- Text, examples or visuals.
- Step-by-step explanations.
- Study plans.
- Verification reminders.
- Adjustable quiz length/level.
- Progress tracking.
- Explain why the next topic is recommended.

Design Insight

The journey suggests that EduGenie should reduce the effort needed to decide what to ask, while keeping the student in control of the learning process and encouraging verification for important facts.

2. REQUIREMENTS ANALYSIS – Problem-Solution Fit

Vision

Make learning simple, personalized and confidence-building using AI.

12. Customer Segments

- Students clearing difficult doubts.
- Students learning from basics to advanced.
- Students revising for exams.

13. Jobs / Problems

- Understand difficult topics (PS-1).
- Know what to learn first/next (PS-2).
- Revise and practice (PS-3).
- Find learning gaps.
- Plan study time.
- Track progress.

14. Triggers

- Difficult topic not understood.
- Starting a subject from scratch.
- Upcoming exams/deadlines.

15. Emotions

Before: confused, unsure, unprepared.

After: clear, confident, in control.

Requirements Analysis – Current Solutions & Behaviour

16. Available Solutions

- Online explanations.
- Scattered resources.
- Self-made questions.
- Basic Q&A tools.

17. Constraints

- Technical explanations.
- No organized learning sequence.
- Revision/question-making takes time.
- Limited study time.

18. Behaviour

- Search online.
- Plan topics independently.
- Re-read material.
- Create own questions.
- Self-check understanding.

19. Channels

- Online: web search, digital material → EduGenie.
- Offline: textbooks, own notes → use EduGenie alongside.

Requirement Insight

The solution should fit existing study habits instead of requiring students to replace textbooks, notes or other learning resources.

Requirements Analysis – Root Causes & Solution

20. Root Causes

- No personalization.
- No learning sequence/study plan.
- No ready practice questions.
- Learning gaps/progress not visible.
- Generic help ignores context.

21. Solution

EduGenie is a Gemini-powered adaptive learning assistant providing simple explanations, roadmap and study plan, adaptive gap checks, progress tracking, revision points, practice questions and context-aware chat.

Key Solution Characteristics

- Mobile-friendly interface.
- One place for multiple learning tasks.
- Learning level can be selected.
- AI output is presented as a study aid and important facts should be verified.

Expected User Value

Students can move from understanding a concept to practicing it and deciding what to learn next without switching between multiple study tools.

Functional & Non-Functional Requirements

Functional Requirements

1. FR-1: Diagnostic & Gap Detection
2. FR-2: Study Plan Generation
3. FR-3: Multi-Format Learning
4. FR-4: Progress & Goal Monitoring
5. FR-5: Context-Aware Support

Non-Functional Requirements

- Usability
- Security
- Reliability
- Performance
- Availability
- Scalability

User Stories – Student

- Explain
- Q&A
- Summary
- Quiz
- Recommendations
- Web UI
- Error handling

Administrator / Developer

- Secure API key using an environment variable.
- Run application locally using Uvicorn.

Acceptance Direction

The system should provide clear task selection, validate input, return a useful response, handle errors, and keep the API key outside the source code.

Key Solution Details, Novelty & Impact

Key Solution Details

Problem	Solution
Students cannot identify what they know/missed.	Gemini-powered AI learning assistant; adaptive diagnosis.
Hard to plan study time.	Personalized study plan and learning roadmap.
Difficult to track progress.	Progress dashboard / progress-oriented support.
Resources are generic/scattered.	Context-aware help, revision and practice.

Novelty

Diagnosis + planning + learning + tracking in one tool. The intended flow is: Diagnose → Plan → Learn → Practice → Track → Re-plan.

Social Impact

- Affordable guidance.
- Saves time.
- Builds self-learning habits.
- Supports confidence.

Business Model

- Freemium.
- Premium subscription.
- Institutional licensing.
- Content partnerships.

Scalability

- Cloud-based.
- Multiple subjects/grades.
- Web/mobile.
- More languages.
- School/college platform integration.

3. PROJECT PLANNING – Sprint Plan

Sprint Plan

Sprint 1

- Explain
- Q&A
- Web UI
- API key security
- Local setup

Sprint 2

- Summarization
- Quiz

Sprint 3

- Recommendations
- Error messages

Sprint Schedule

The documentation uses a consistent project date of 2 October 2026. Sprint 1 covers the base application and core interaction; Sprint 2 adds revision and practice functions; Sprint 3 adds recommendations and robustness.

Planning Principle

The plan moves from core infrastructure to learning features and then to recommendation/error-handling improvements.

Project Planning – Product Backlog & Team

Product Backlog

11 stories – 32 points

Sprint	Story	Points
Sprint 1	USN-1: FastAPI + Uvicorn backend	2
Sprint 1	USN-2: HTML/CSS/Jinja2 UI	3
Sprint 1	USN-3: Local LaMini model /explain	3
Sprint 1	USN-4: Gemini API	2
Sprint 1	USN-5: Error handling	2
Sprint 2	USN-6: Q&A	2
Sprint 2	USN-7: Quiz	3
Sprint 2	USN-8: Summary	2
Sprint 2	USN-9: Learning path	3
Sprint 2	USN-10: Jinja2 rendering	5
Sprint 2	USN-11: Documentation and demo	5

Team Members

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Project Planning – Velocity, Burndown & Priority

Velocity

Total = 32 points. $32 \div 2 = 16$ points/sprint. $16 \div 6 \approx 2.67$ points/day.

Burndown Chart

Shows work remaining vs time. Planned: $32 \rightarrow 20 \rightarrow 0$. Actual progress can be added later.

Burndown (planned): $32 \rightarrow 20 \rightarrow 0$

Priority

- High: USN-1, 2, 3, 4, 6, 10
- Medium: USN-5, 7, 8, 9, 11

Tracker

- Story points completed.
- Actual sprint release date.

4. PROJECT DESIGN PHASE – Technical Architecture

Proposed Technical Architecture

Architecture Flow

Student	HTML/CSS UI	FastAPI	Feature Module	AI Model	Response	Jinja2	UI
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Architecture Layers

Layer	Components
Client	Student / Learner
Presentation	HTML, CSS, Jinja2
Application	FastAPI, Python 3.10+, Uvicorn
AI Processing	Local AI model, Gemini Cloud API

Design Goal

Keep the interface simple while separating UI, API routing, feature modules and AI processing so that each part can be maintained independently.

Technology Stack & Application Modules

Technology Stack

Area	Technology
Frontend	HTML, CSS
Backend	FastAPI
Server	Uvicorn
Template	Jinja2
Storage	Local filesystem
Database	None

Application Modules

- main.py
- explanation_module.py
- qna.py
- quiz_module.py
- summary_module.py
- learning_path.py

AI Models

Local

LaMini-Flan-T5-783M → /explain

Cloud

Google Gemini → /qa, /quiz, /summarize, /learn/recommendations

Model Configuration

The Gemini model is configurable through GEMINI_MODEL. The example configuration uses gemini-2.5-flash; availability can change and the model should be set to one enabled for the API key.

Data Flow Diagram & Data Handling

DFD Flow

Student enters topic / question / study text	FastAPI / Uvicorn routes request	/explain → Local LaMini model	Other features → Gemini API	Jinja2 renders response	Student sees output
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Data Storage

- No user data stored.
- Local template files only.

DFD Processes

1. 1.0 Handle Request
2. 2.0 Explain Concept
3. 3.0 Answer Questions
4. 4.0 Generate Quiz
5. 5.0 Summarize Content
6. 6.0 Learning Recommendations
7. 7.0 Render Response

API Endpoints

- GET /
- GET /health
- POST /qa
- POST /explain
- POST /quiz
- POST /summarize
- POST /learn/recommendations

Application Characteristics & Recommended Improvements

Application Characteristics

- Open source technologies.
- Backend API security.
- No user data stored.
- Modular architecture.
- Async design.
- /explain can work offline.
- Gemini features require internet.
- Local model avoids network call.

Recommended Improvements

- Pin dependencies.
- Vulnerability scanning.
- Environment variables / secret manager.
- HTTPS.
- Input validation.
- Rate limiting.
- Authentication.
- Docker.
- Multiple workers.
- Load balancer.
- Cloud deployment.
- Health checks.
- Retry / fallback.
- Response caching.
- GPU / quantization.
- Latency monitoring.

Security Note

The Gemini API key is read from an environment variable and should not be committed or shared. Production deployment should add stronger secret management, HTTPS, authentication and rate limiting.

Implementation – Web UI & FastAPI

HTML Interface

The web page provides a task dropdown, learning-level dropdown, text area, Generate answer button and result panel. Tasks include Explain a concept, Ask a question, Generate a quiz, Summarize notes and Create a learning path.

Style.css

The supplied stylesheet uses a responsive centered shell, panel/output cards, rounded controls, focus states, quiz-choice feedback and a compact mobile layout. The project documentation keeps the supplied style concept without changing the documentation template.

Main.py – Core Flow

```
app = FastAPI(title="EduGenie", version="1.0.0")

@app.get("/", response_class=HTMLResponse)
async def home(request: Request):
    return templates.TemplateResponse("index.html", {"request": request})

@app.post("/explain")
async def explain(payload: TextRequest):
    return {"result": await explain_concept(clean_input(payload.text), payload.level)}
```

Health Check

```
@app.get("/health")
async def health():
    return {"status": "ok", "service": "EduGenie"}
```

Implementation – AI Client, Features & Testing

AI Client

```
key = os.getenv("GEMINI_API_KEY", "").strip()
if not key:
    raise RuntimeError("GEMINI_API_KEY is missing.")
return genai.Client(api_key=key)
```

Explanation Module

`explain_concept()` creates a learner-level prompt. When `USE_LOCAL_EXPLAINER=true`, LaMini-Flan-T5-783M is used; if the local model fails and `LOCAL_FALLBACK_TO_GEMINI=true`, the function falls back to Gemini.

Learning Path

`get_learning_recommendations()` asks Gemini to organize a topic from beginner to intermediate to advanced, including ordered topics, time estimates, practice projects and resource types.

Quiz Module

`generate_quiz()` requests exactly three MCQs with four options each, a correct answer and a brief explanation. `clean_json_block()` extracts JSON from plain or fenced model output.

Testing

```
python -m unittest discover -s tests -v
```

The supplied tests validate quiz JSON cleanup for plain JSON, Markdown-fenced JSON and JSON surrounded by text.

Implementation – README, Requirements & Conclusion

README – Setup

1. Install Python 3.10 or newer and pip.
2. Create a virtual environment.
3. Install requirements with `pip install -r requirements.txt`.
4. Create `.env` and add `GEMINI_API_KEY`.
5. Start the server using: `uvicorn main:app --reload`.
6. Open `http://127.0.0.1:8000` and `/docs` for API documentation.

Requirements

```
fastapi>=0.115,<1
uvicorn[standard]>=0.30,<1
jinja2>=3.1,<4
google-generativeai>=1.0,<2
python-dotenv>=1.0,<2
pydantic>=2.7,<3
```

Optional Local Model

`torch>=2.2` and `transformers>=4.40,<5` are needed only when `USE_LOCAL_EXPLAINER=true`. The model downloads on first use and may require substantial disk/RAM.

Conclusion

EduGenie combines AI-powered explanation, Q&A, summarization, quiz generation and learning recommendations in one student-friendly assistant. Its modular FastAPI architecture separates presentation, application logic and AI processing, while environment-based API key handling supports safer development.

The project is designed to help students understand difficult topics, revise efficiently, practise and follow a logical learning path. AI output remains a study aid and important information should be verified.